chapter 3

Central tendency - Central Location

Mean: $X_1 \dots X_n$ prone to outliers

- population mean μ (unknown) parameter
- sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$ random variable

random variable means that can change at different samplings.

$E[X] = \mu$ (unbiased) mean is unbiased estimator.

Median: insensitive to outliers (extreme values)

sorted $X_1^s \dots X_{2n-1}^s$ $n \in \mathbb{Z}$ $X_1^s \dots X_{2n}^s$

X_n $\frac{X_n + X_{n+1}}{2}$

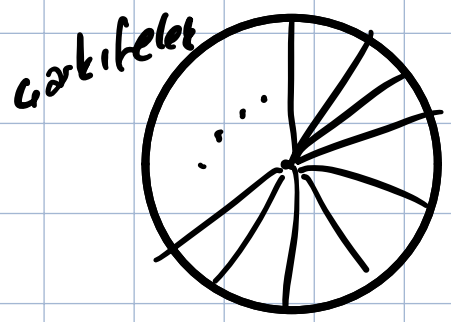
Mode: most observed value

categorical data

Expected Value: mean is a special case that every occurence has equal probability.

$$EV = \sum_{i=1}^{n} X_i p_i \qquad n \text{ possible outcomes}$$

Ex: \$5 ticket



1	\$50
5	\$15
10	\$5
34	\$0

$$\frac{45 + 50 + 0 - 170}{50} = -\$1.50 \sim$$

$$EV = \frac{1}{50} \cdot 45 + \frac{5}{50} \cdot 10 + \frac{10}{50} \cdot 0 + \frac{34}{50}(-5)$$

proportions for binary data

$$\text{e.g.} = \frac{\#\text{ of 1s}}{\#\text{ of 0s and 1s}}$$

Variability

Range: $X_1 \ldots X_n$ $\quad \max_{i=1\ldots n} X_i - \min_{i=1\ldots n} X_i$

single value

Percentiles: $X_1 \ldots X_n \xrightarrow{\text{sort}} X_1^s \ldots X_n^s$ sorted

%25

%75

100 sayı varsa %75 dendiğinde 75. sayı 75 percentile olur.

50 %ile is the median

percentile

Interquartile range: $X_{75\%} - X_{25\%}$

single value.

Deviations & Residuals:

Residual: mean value − current value $x_i - \bar{x}$

Ex: 1, 4, 4 mean: $3 = \bar{x}$

$1 - 3 = -2$
$4 - 3 = 1$
$4 - 3 = 1$

residuals

deviations from the mean

Mean absolute deviation:

$$\frac{1}{n} \sum_{i=1}^{n} |x_i - \bar{x}|$$

$$\frac{2+1+1}{3} = \frac{4}{3}$$

Variance and Standard Deviation

MAD $\frac{1}{n}\sum_{i=1}^{n}|x_i-\bar{x}|$

mean abs. dev.

kasey: unutursat variance O cikar.

Variance: $\sigma^2=\frac{1}{n}\sum(x_i-\mu)^2$ σ^2: variance of population

dof: $n-1$

degrees of freedom

1 to find $\bar{x}$

$$\boxed{s^2=\frac{1}{n-1}\sum(x_i-\bar{x})^2}$$

s^2: sample variance

unbiased estimator

$E[s^2]=\sigma^2$ if this, s^2 is unbiased estimator

Euclidian distance:

$$\sqrt{(w_1-x_1)^2+\ldots(w_n-x_n)^2}$$

Person	Rock	Hip-Hop	Country	Jazz	New Age
A	7	1	9	1	3
B	4	9	1	3	1
C	9	1	7	2	2

Definition: Vector A vector is a row of numbers. Vector arithmetic is done by performing the operation on the corresponding elements of each vector, resulting in a new vector of sums, differences, products, etc.

The statistic that we have described is "Euclidean Distance." Here is the formula, followed by the calculations.

As a general example, assume we have two vectors, w and x, each containing n values. The Euclidean Distance between the two vectors is

$$\text{Euclidean Distance} = \sqrt{(w_1-x_1)^2+(w_2-x_2)^2+\cdots(w_n-x_n)^2}$$

If you look carefully at the formula, you might recognize that this is the multidimensional version of the formula for the distance between two points that you may have learned in geometry.

For a specific example, the Euclidean Distance between vectors A and B—a measure of how alike person A is to person B—in Table 3.2 is calculated in Table 3.3.

In Table 3.3, the sum of the squares of the differences of each row is 145. The square root of 145 is 12.04, which is the Euclidean Distance between the two vectors representing person A and person B. Looking back at the data in Table 3.2, try the following problem:

Test Statistics

Hypothesis Testing

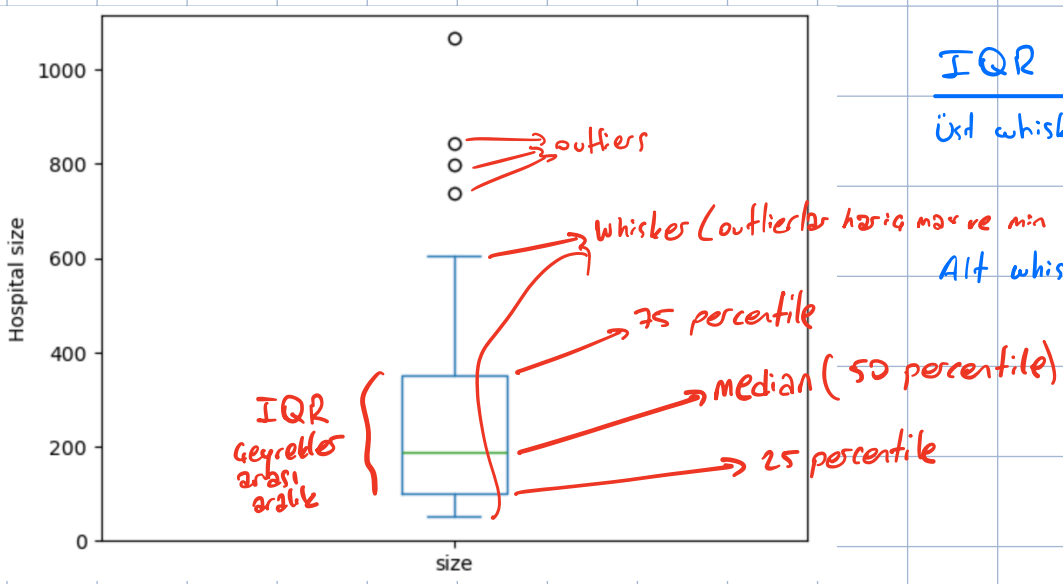
Hospital Error Reduction

Control mean reduction $\bar{X}_c$

Treatmen mean reduction $\bar{X}_t$

$$t_1 = \bar{X}_t - \bar{X}_c$$

$$t_2 = \frac{X_t}{\bar{X}_c} - 1$$



IQR hesabı (whisker)

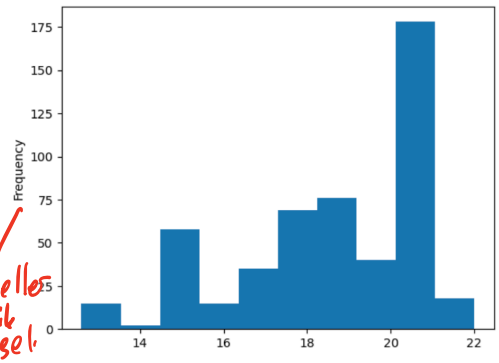
Üst whisker: $Q_3 + 1{,}5\,IQR$ içerisindeki en büyük değer

Alt whisker: $Q_1 - 1{,}5\,IQR$ içindeki en küçük değer

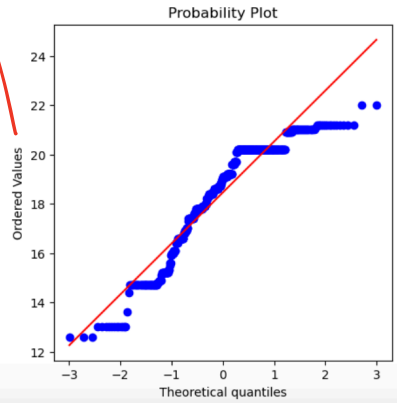
```
[6]: from scipy import stats

housing = pd.read_csv("boston-housing-large.csv")
housing["PTRATIO"].plot.hist(); #sona noktalı virgül run edildiğinde print edilmemesini sağlıyormuş
fig,ax = plt.subplots(figsize=(5,5))
stats.probplot(housing["PTRATIO"], plot=ax, dist="norm");
```

• histogram bir tarafa yatıksa normal dağılımdan sapmalar vardır.



bu labeller otomatik geli.



histogramdan daha hassas

kırmızı: ideal normal dağılım

mavi: .csv verileri

Chapter 4

Hypothesis testing - Permutation test

Test statistic $\bar{X}_t - \bar{X}_c$

We want to check if $\bar{X}_t > \bar{X}_c$?

X_t: error reduction after treatment

X_c: error reduction in control group

H_0 null hypothesis $\bar{X}_t = \bar{X}_c$ | $\bar{X}_c \geq \bar{X}_t$

H_1 "alternative" hypothesis (not null, not equal) $\bar{X}_t \neq \bar{X}_c$ | $\bar{X}_t > \bar{X}_c$

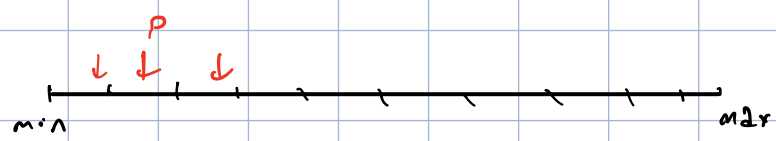
We would like them to be $N(\mu_t, \sigma_t^2)$, in that case, you may apply t-test.

$N(\mu_c, \sigma_c^2)$

or least $n \geq 30$ — CLT kicks in. (parametric test)

If not, permutation test. (non-parametric test)

$C\binom{50}{25}$ diffent possibilities (exact test)



p is the probability of getting observed value or more

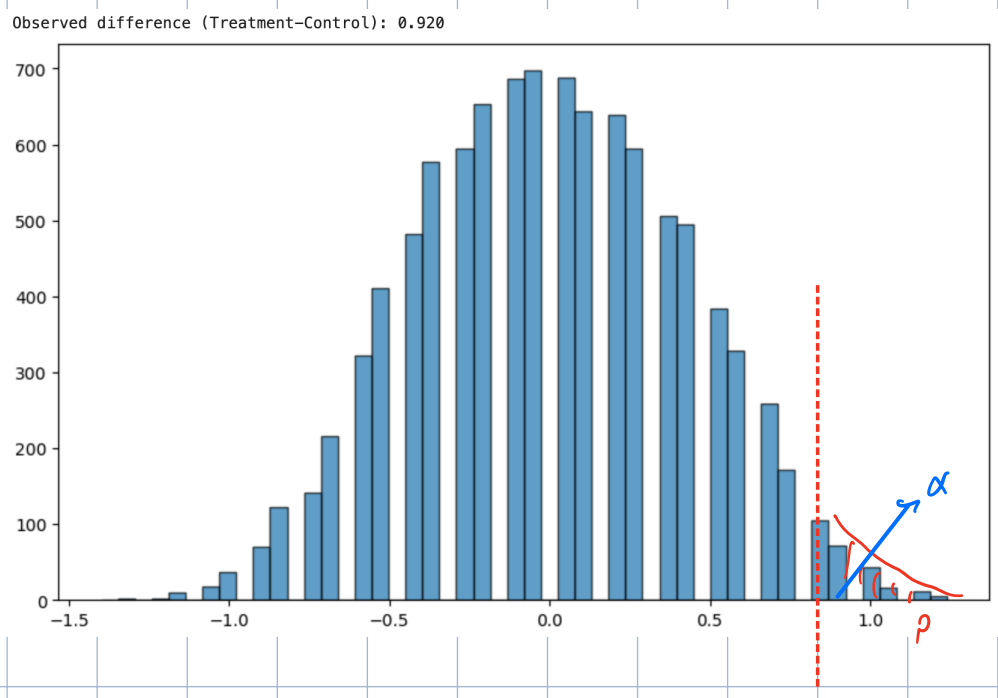
sınav sorusu

α : Type-1 error probability

$\alpha = 0.05$ if $\alpha > p \Rightarrow$ reject the null hypothesis

hiçbir zaman accept kullanma. not reject kullan

Type-1 error: rejecting an H_0 (null hyp.) when you should not have.



Type-2 error: not rejecting H_0 when you should have.

β : type-2 error probability

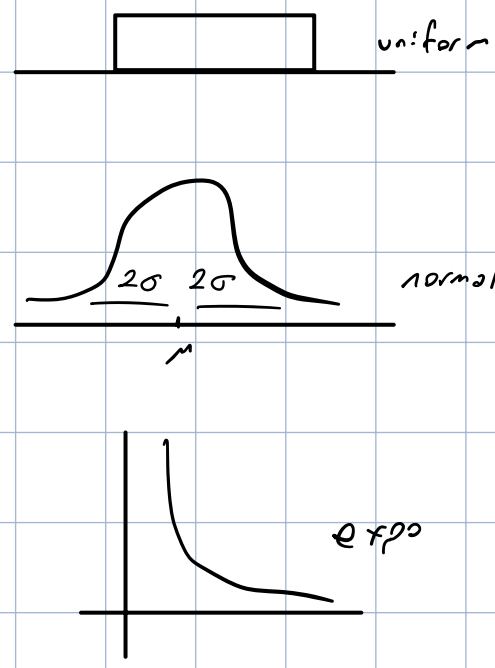
1. Sıfır Hipotezi (H_0) Nedir?

- **Tanım:** Bir deneyin başında kabul edilen "varsayılan" durumdur.
- **Mantık:** Genellikle "gruplar arasında fark yoktur" veya "tedavinin bir etkisi yoktur" anlamına gelir.
- **Örnek:** Notlarındaki hastane örneğinde H_0, "Yeni hata azaltma yöntemi ile eski yöntem arasında bir fark yoktur" demektir.

2. Tip-1 Hata (Yalancı Pozitif)

Tip-1 hata, **Sıfır Hipotezi aslında doğruyken** (yani gerçekte bir fark yokken), yanlışlıkla onu reddedip "bir fark var" dediğimizde oluşur.

- **İlişki:** H_0 doğru olduğu halde, örneklemden elde edilen verilerin şans eseri uç değerlerde çıkması sonucu araştırmacının "H0 yanlıştır" sonucuna varmasıdır.
- **Yanılgı:** Gerçekte hiçbir etkisi olmayan bir ilacın veya yöntemin "işe yarıyor" gibi görünmesidir.
- **Anlamlılık Düzeyi (α):** Tip-1 hata yapma olasılığına α denir. Genellikle bu oran %5 (0.05) olarak belirlenir. Bu, "100 testten 5'inde aslında fark yokken varmış gibi hata yapabilirim" demektir.



Chapter 5

- Binomial distribution (Discrete) PMF, CDF

  n independent trials < success / failure

  $Prob\{x=k\}$ x: # of successes in n trials

  $$= \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

- Multinomial distribution farklı örnekler varsa kullanılır, binomial'da gösterilen her şey burada da gösterilir ama bu advance.

- De Moivre - Laplace Theorem

  As $n \to \infty$, for a constant p

  $$X \overset{\text{distributed}}{\sim} Binomial(n,p) \xrightarrow[\text{converges}]{} N(\mu, \sigma^2)$$

  $\mu = n.p$, $\sigma^2 = n.p.(1-p)$

Standarding to a Z-score

$$Z = \frac{X-\mu}{\sigma}$$

$X \sim N(\mu, \sigma^2)$

$Z \sim N(0,1)$: standard, normal

$$Z = \frac{X - np}{\sqrt{np(1-p)}}$$

n large, then use Z-table for $Pr\{X \leq k\}$

$$Z^2 = \frac{(X-n.p)^2}{n.p.(1-p)}$$

$Z \sim N(0,1) \Rightarrow Z^2 \sim \chi^2$ → x değil, chi

$$Z^2 = \frac{(x-np)^2}{n} \cdot \left(\frac{1}{p} + \frac{1}{1-p}\right) = \frac{(x-np)^2}{np} + \frac{(x-np)^2}{n(1-p)}$$

(+n ve -n eklendi)

$$= \frac{(x-np)^2}{np} + \frac{((n-x) - n(1-p))^2}{n(1-p)}$$

Two categories — success (p), failure (1-p)

$$= \frac{(O_s - E_s)^2}{E_s} + \frac{(O_f - E_f)^2}{E_f} \sim \chi_1^2$$

Pearsons's formula

$E_s = n.p$

$E_f = n.(1-p)$

$O_s = x$ → observed.

$O_f = n - x$

Similarly,

$$\sum_{i=1}^{m} \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{m-1}$$

← degrees of freedom

Chi-square test

standard normal dağılımın kare si

Example

$n = 100$, $p = 0,3$

At least 40 success

using this table, calculate

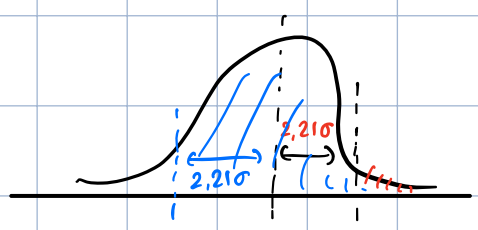
$$\binom{n}{k} p^k (1-p)^k ? = z = \frac{x - \mu}{\sigma}$$

mean $n.p = 30$

var. $n.p.(1-p) = 21$

$$\frac{(40 - 30)}{\sqrt{21}} = \frac{10}{\sqrt{21}} = 2,21$$

tablodan 0,9864



at least: (1-0,9864) probability = 1,36 %

Z-Score Table (Standard Normal Table)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

At least 20 success

$$\frac{(20 - 30)}{\sqrt{21}} = -2,21$$

at least: 0,9864

= 98,64 %

Chi-Square table

Degrees of freedom (df)	Significance level (α) .99	.975	.95	.9	.1	.05	.025	.01
1	-------	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932
22	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980
25	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314
26	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642
27	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963
28	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278
29	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588
30	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892
40	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691
50	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154
60	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379
70	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425
80	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329
100	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116
1000	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807

Chapter 6

Conditional Probability

$$Pr\{X = x\}$$

X : # of goals scored

$x = 0$	30 %
$x = 1$	45 %
$x = 2$	20 %
$x = 3+$	5 %

Y : information (Osimhen not playing)

$Pr\{X = x \mid Y\}$ prob of $X = x$ given Y

$x = 0$	60 %
$x = 1$	35 %
$x = 2$	5 %

$Pr(\text{Acceptance to } X \mid \text{female})$

• Berkley ödevi: HW
chi square test

Consider a medical screening test that gives a positive result in 98% of the cases where the condition is present but also gives a false positive result in 3% of cases where the condition is not present. Suppose 0.1% of the people we screen actually have the condition (to pick an actual disease, this is the approximate prevalence of juvenile arthritis).

If you test positive, what is the probability that you have the disease?

X: outcome

H : health cond.

given

$Pr\{X = 1 \mid H = 1\} = 98\%$

$Pr\{X = 1 \mid H = 0\} = 3\%$

$Pr\{H = 1\} = 0{,}1\%$

question

$Pr\{H = 1 \mid X = 1\} = ?$

```
"CRIM_RM"]])
```

```
n Results
==============================================
-squared:                                0.491
dj. R-squared:                           0.485
-statistic:                              84.43
rob (F-statistic):                    2.74e-38
og-Likelihood:                        -702.63
IC:                                     1413.
IC:                                     1428.

==============================================
 t      P>|t|      [0.025      0.975]
----------------------------------------------
80      0.030     -14.510      -0.737
90      0.001       0.501       1.993
63      0.000       3.551       5.809
07      0.000      -0.416      -0.167
==============================================
urbin-Watson:                            1.860
arque-Bera (JB):                        38.218
rob(JB):                              5.02e-09
ond. No.                                  650.
```

$$\} = \frac{Pr(A \cap B)}{P(B)}, \quad \boxed{Pr(B) \cdot Pr(A \mid B) = Pr(A \cap B) = Pr(A) \cdot Pr(A \mid B)}$$

==========

iance matrix of the errors is correctly specified.

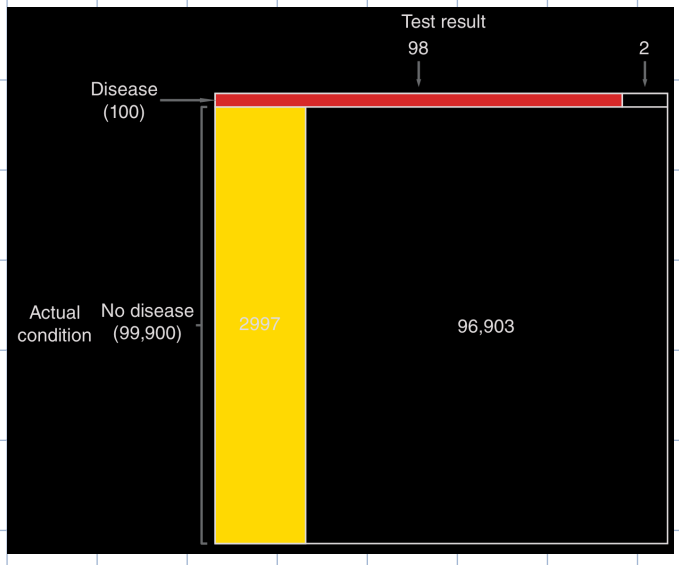
$$\boxed{Pr(A) = Pr(A|B).Pr(B) + Pr(A|B^c).Pr(B^c)}$$

$$Pr(H=1|X=1) = \frac{Pr(H=1, X=1)}{Pr(X=1)}$$

$$= \frac{Pr(H=1, X=1)}{Pr(X=1|H=1).Pr(H=1) + Pr(X=1|H=0).Pr(H=0)}$$

$$= \frac{Pr(X=1|H=1).Pr(H=1)}{Pr(X=1|H=1).Pr(H=1) + Pr(X=1|H=0).Pr(H=0)}$$

$$= \frac{0,98 \cdot 0,001}{0,98 \cdot 0,001 + 0,09 \cdot 0,999} = 0,0316$$



Bayesian calculation

$$Pr = \frac{98}{98 + 2997} = 0,0316$$

Independence

$$Pr(X=x|Y) = Pr(X=x)$$

Chi square test (Hypothesis testing)

Does the calibration requirement depend on the state?

H_0: recalibration does not depend on the state.

	+	−	
TX	25 (a)	175	200-a
CA	17 (42-a)	183	200-42+a

$\boxed{421400}$

observed

	+	−
TX	21	179
CA	21	179

expected (null hyp.)

$$\sum_{i=1}^{4} \frac{(O_i - E_i)^2}{E_i} \sim \chi_i^2$$

bu chi2 tablosunda E_i hiçbir değer 5'ten küçük olamaz. ≥ 5

dof = 1, değiştirebileceğimiz tek variable var

dof = (r-1)(c-1)

{row-1, column-1}

$$= \frac{(25-21)^2}{21} + \frac{(17-21)^2}{21} + \frac{(175-179)^2}{179} + \frac{(183-175)^2}{175} = 1.703$$

chi-sq tableda dof=1 için 0,05 değer: $\chi^2_{1,0.05} = 3.841$

$\chi_0^2 = 1.703 < \chi^2_{1,0.05} = 3.841$

do not reject H_0

* chi-sq test kesin sınav sorusu
* sınava chi2 ve z table çıktısını al, hesap makinesi al.
* multiple choice : what is p value gibi
* hand calculations 2 or 3 questions
* coding 2 or 3
* ders notu ve kitaptan ch2 - ch6 çıktısını al

t-tests: / alternative to permutation test

hypothesis testing for

- unknown variance $\longrightarrow$ σ is known z-test
- small number of samples

conditions:

- numeric and continuous (blood pressure, weight, exam scores)
- independence of observations
- $\underbrace{\text{approximate normality}}_{\text{Q-Q plot}}$ or large sample ($n \geq 30$)

t-tests

one sample (e.g. mean) — two samples (comparison of means) independent samples

paired t-test $d_i = x_i - y_i$

: t has two samples, but is solved like one sample. (take difference of two samples and act like a one sample.)

- one sample test $X_j \sim N$
- two sample test $\left.\begin{array}{l} X_i \sim N(\mu_i, \sigma_i) \\ Y_j \sim N(\mu_j, \sigma_j) \end{array}\right\}$ paired t-test $d_i = x_i, y_i \sim N(,)$

t-test

one-sided — two-sided

One-sample two-sided test

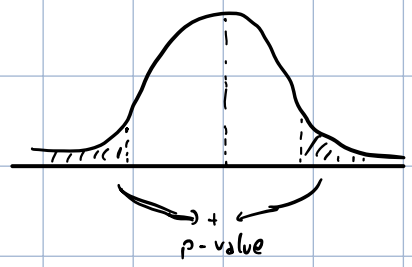
$H_0: \mu = \mu_0$

$H_1: \mu \neq \mu_0$

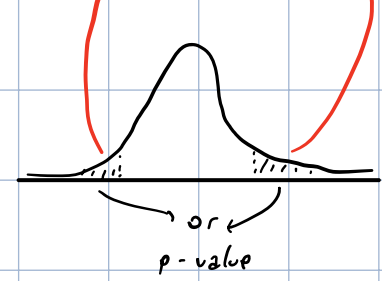
One-sample one-sided test

$H_0: \mu \geq \mu_0$ or $\mu \leq \mu_0$

$H_1: \mu < \mu_0$ $\quad$ $\mu > \mu_0$



$|t|$ vs. $t_{\alpha/2, n-1}$



t vs. $\mp t_{\alpha, n-1}$

test statistic

$x_1, x_2, \ldots, x_n \sim N(\mu, \sigma)$

mean: $\bar{x}$, s

$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i \qquad s = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2}$

t-statistic

$$\boxed{t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}}$$

r : dof $\quad v = n-1$

two-sided

Ex: $\mu_0 = 500$ gr

$n = 12$ bags

data: 483, 471, 478, 465
460, 463, 474, 458
472, 466, 461, 475

$H_0: \mu = 500$

$H_1: \mu \neq 500$

$\alpha = 0{,}005$

$\bar{x} = 469.33$ gr

$s = 7{,}63$ gr

$$t = \frac{469.33 - 500}{7{,}63/\sqrt{12}} = \frac{-30{,}67}{2{,}20} = -13{,}93$$

t-table

$t_{\alpha/2, n-1} = t_{0{,}025, 11} = 2{,}201$

because two-sided

$$\boxed{\text{since } |-13{,}93| \geq 2{,}201 \text{ reject } H_0}$$

Two sample (welch) t-test

$x_i \sim N(\mu_x, \sigma_x)$ n_1

$y_i \sim N(\mu_y, \sigma_y)$ n_2

two-sided

- $H_0: \mu_1 = \mu_2$
- $H_1: \mu_1 \neq \mu_2$

one-sided

- $H_0: \mu_1 \geq \mu_2$
- $H_1: \mu_1 < \mu_2$

- $H_0: \mu_1 \leq \mu_2$
- $H_0: \mu_1 > \mu_2$

Ex: Octane booster

H_0: octane booster does not improve 0-100 km/h performance : different sets of cars

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{s_x^2}{n_1} + \frac{s_y^2}{n_2}}}$$

d.o.f.

$$v = \left\lfloor \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} \right\rfloor$$

→ floor of this value.

Paired t-test

multi-commodity

MC: $x_1, x_2, \ldots, x_n$

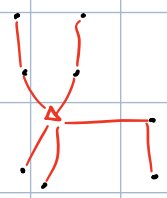
SC: $y_1, y_2, \ldots, y_n$

Single-commodity

optimality gaps

lower is the better

the same number of samples



facility location, network design

multi-commodity 3-indices d.v

single-commodity 2-indices dec.var.

For 30 nodes, 50 random setup (instances)

$d_i = x_i - y_i \quad i = 1, 2, \ldots, n$

$\underbrace{\text{Check if } d \sim N(\,,\,)}_{\text{Q-Q plot}}$ or $n \geq 30$

$\bar{d}, s_d$

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

One Tail T-Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

Ex: Solution times

MC:	15.2	14.8	16.5	15.0	15.9	14.5	16.2	15.3	14.7	16.0
SC:	12.5	13.0	12.8	13.5	12.2	13.1	12.7	12.9	13.3	12.4
d:	2.7	1.8	3.7	1.5	3.7	1.4	3.5	2.4	1.4	3.6

$H_0: \mu_d = 0$

$H_1: \mu_d \neq 0$

$\alpha = 0.05$

$\bar{d} = 2.57$ sec

$s_d = 1.00$ sec

$$t = \frac{2.57}{1/\sqrt{10}} = 8.13 \qquad \text{dof } v = 9$$

$t_{\alpha/2, 9} = 2.262$

$$\boxed{\text{Since } |t| = 8.13 \geq t_{0.025, 9} = 2.262 \text{ reject } H_0}$$

Confidence Intervals & Bootstrapping for the mean

$X_1, \ldots, X_n \sim N$ or $n \geq 30$

$\bar{x}$ point estimate

Confidence Interval CI $\bar{x} - t_{\alpha/2, n-1}(s.e)$ — $\bar{x}$ — $\bar{x} + t_{\alpha/2, n-1}.(s.e.)$ for $(1-\alpha)$ confidence

s.e : standard error $= \frac{s}{\sqrt{n}}$

If these are not satisfied : Bootstrapping kesin sınav sorusu

---

Table 8.1 Marriage therapy. observed values O_i

	Happily Married	Distress Married	Divorced	
Behavioral	15	3	11	29
Insight	24	5	1	30
	39	8	12	59

H_0 : The marriage status does not depend on the therapy type

$E_i = \frac{12}{59} \cdot 29$

$E_i = \frac{39}{59} \cdot 29$

$E_i = \frac{8}{59} \cdot 29$

$E_i = \frac{12}{59} \cdot 30$

$E_i = \frac{39}{59} \cdot 30$

$E_i = \frac{8}{59} \cdot 30$

Expected table

Table 8.2 Expected outcomes if treatments yield the same results.

	Happily Married	Distress Married	Divorced
Behavioral	19.17	3.93	5.90
Insight	19.83	4.07	6.10

>5, so okay

not okay since chi sq test requires >5

then, we need to combine distress married data with smt.

Table 8.3 Absolute difference between observed and expected.

	Happily Married	Distress Married	Divorced
Behavioral	4.17	0.93	5.11
Insight	4.17	0.93	5.11

Goodness of Fit

H_0: These numbers are samples uniformly. equal prob of $1/10$ for selection

315 samples

Digit	Observed Frequency	Expected	Diff	Diff²/E
0	14	31.5	17,5	9,72
1	71	31.5	39,5	49,53
2	7	31.5	24,5	19,05
3	65	31,5	33,5	35,63
4	23	31,5	8,5	2,29
5	19	31,5	12,5	4,96
6	12	31,5	19,5	12,07
7	45	31,5	13,5	5,78
8	53	31.5	21,5	14,67
9	6	31,5	25,5	20,64
			+	174,34

$dof = 10 - 1 = 9$

from chi-square table: $\chi_{0,05,9} = 16,919$ so, reject H_0.

	Fat 1	Fat 2	Fat 3	Fat 4
Replication 1	164	178	175	155
Replication 2	172	191	193	166
Replication 3	168	197	178	149
Replication 4	177	182	171	164
Replication 5	156	185	163	170
Replication 6	195	177	176	168
Average	172	185	176	162

we need to compare 1 by 1.

H_0: they have the same mean

$(1-\alpha)$: prob of not making an error.

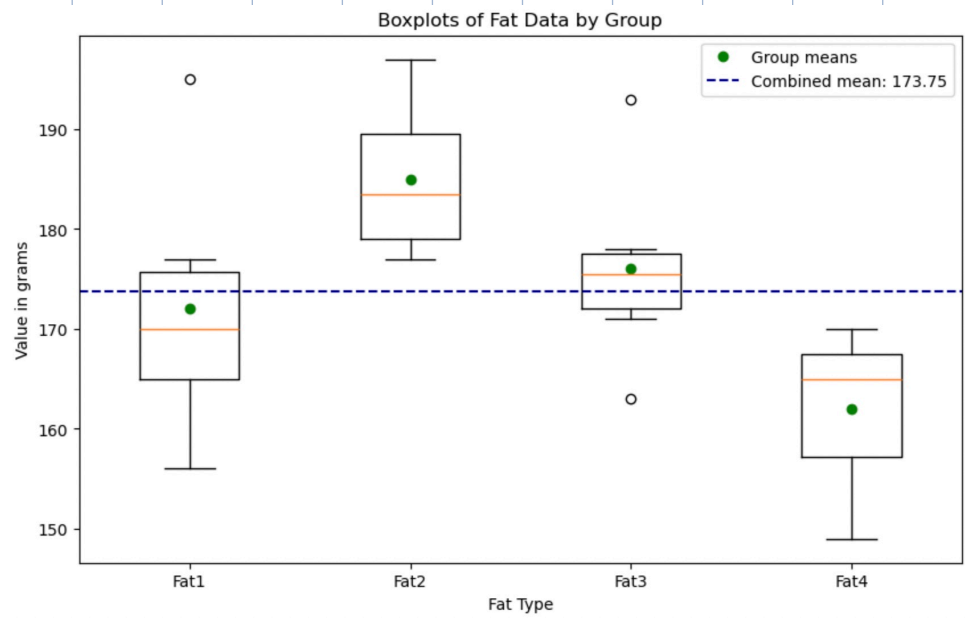
$(1-\alpha).(1-\alpha).(1-\alpha)$ 3 test yapıldığı için

showing that fat4 has a higher Type-1 prob than α.

Eğer $\alpha = 0,05$ seçilirse küpü 0,143 olur. ÇOK BÜYÜK!!!

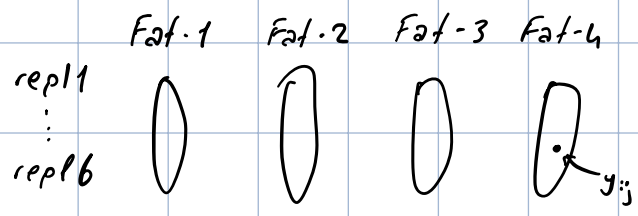
Bu yüzden α başta daha küçük seçilmeli.

ANOVA : Analysis of Variance



Analysis of Variance

One-way ANOVA



$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$

$\binom{4}{2} = 6$

α_r : Type-1 error $\quad 1-(1-\alpha_r)^6$: Type-1 error

- If you have the need to run multiple t-tests (3t) at once, that will inflate the type-1 error probability.

ANOVA solution

Compute the distinct type of variances
- Between group variance "signal"
- Within group variance "noise"

Logic: If signal > noise, the group means μ_i are different

Assumptions:
- Independence
- Normality
- Variances are roughly equal

$$y_{ij} = \mu + \underbrace{\tau_i}_{\text{signal}} + \underbrace{\epsilon_{ij}}_{\text{noise}}$$

μ : overall mean → grand mean

τ_i : correction term for group; avg. ($\mu + \tau_i$: avg of ith group) → treatment effect

ϵ_{ij} : deviation y_{ij} from the group mean $\mu + \tau_i$ → random error

$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$

H_1: at least one group mean is different.

Evaluating EV Battery Architecture

Goal: Compare fast-charging times (20% - 80%) of three battery type.

Groups $(k=3)$ LFP, NMC, NCA

Battery Type	Times (s)	Group Mean	Grand Mean
LFP	28, 30, 28, 31	28,5	24,67
NMC	22, 24, 25, 23	23,5	
NCA	20, 21, 23, 20	21,0	

Partition the Variance

SS: Sum of squares

$$SS_{total} = SS_{between} + SS_{within}$$

$$\sum_i \sum_j (y_{ij} - \mu)^2 = \sum_i \sum_j (\mu_i - \mu)^2 + \sum_i \sum_j (y_{ij} - \mu_i)^2$$

$$SS_{between} = 4.(28,5 - 24,67)^2 + 4.(23,5 - 24,67)^2 + 4.(21 - 24,67)^2 = 152,67$$ related to $var(\tau_i)$

$$SS_{within} = (28 - 28,5)^2 + (30 - 28,5)^2 \ldots\ldots = 16$$ related to $var(\epsilon_{ij})$

ANOVA Table (sınavda çıkacak)

variance

Source	SS	dof	Mean Square	F - Statistic
Between	152,67	$k-1=2$	$152,67/2 = 76,33$	$F = 76,33/1,78 = 42,88$
Within	+ 16	+ $N-k=9$	$16/9 = 1,78$	
Total	168,67	$N-1=11$		

The F - Statistic $F = \dfrac{MS_{between}}{MS_{within}} = \dfrac{\text{signal}}{\text{noise}}$ SNR

for $\alpha = 0,05$

From F Table $F_{2,9} = 4,2565$ critical value

$F > F_{2,9}$ reject H_0

Critical Values of the F-Distribution: $\alpha = 0.05$

Denom. d.f.	Numerator Degrees of Freedom 1	2	3	4	5	6	7	8	9	10
1	161.448	199.500	215.707	224.583	230.162	233.986	236.768	238.883	240.543	241.882
2	18.513	19.000	19.164	19.247	19.296	19.330	19.353	19.371	19.385	19.396
3	10.128	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.786
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637
8	5.318	4.459	4.066	3.838	3.687	3.581	3.500	3.438	3.388	3.347
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348
21	4.325	3.467	3.072	2.840	2.685	2.573	2.488	2.420	2.366	2.321
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255
25	4.242	3.385	2.991	2.759	2.603	2.490	2.405	2.337	2.282	2.236
26	4.225	3.369	2.975	2.743	2.587	2.474	2.388	2.321	2.265	2.220
27	4.210	3.354	2.960	2.728	2.572	2.459	2.373	2.305	2.250	2.204
28	4.196	3.340	2.947	2.714	2.558	2.445	2.359	2.291	2.236	2.190
29	4.183	3.328	2.934	2.701	2.545	2.432	2.346	2.278	2.223	2.177
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.211	2.165
31	4.160	3.305	2.911	2.679	2.523	2.409	2.323	2.255	2.199	2.153
32	4.149	3.295	2.901	2.668	2.512	2.399	2.313	2.244	2.189	2.142
33	4.139	3.285	2.892	2.659	2.503	2.389	2.303	2.235	2.179	2.133

Special Case: (Anova 3 veya daha fazla grup olursa kullan).

When comparing exactly two groups, one-way ANOVA and independent two-sample t-test are mathematically identical.

$$F = t^2$$

$$
\underbrace{\begin{pmatrix} 164 & 178 & 175 & 155 \\ 172 & 191 & 193 & 166 \\ 168 & \boxed{197} & 178 & 149 \\ 177 & 182 & 171 & 164 \\ 156 & 185 & 163 & 170 \\ 195 & 177 & 176 & 168 \end{pmatrix}}_{\text{Observations}}
=
\underbrace{\begin{pmatrix} 173.75 & 173.75 & 173.75 & 173.75 \\ 173.75 & 173.75 & 173.75 & 173.75 \\ 173.75 & \boxed{173.75} & 173.75 & 173.75 \\ 173.75 & 173.75 & 173.75 & 173.75 \\ 173.75 & 173.75 & 173.75 & 173.75 \\ 173.75 & 173.75 & 173.75 & 173.75 \end{pmatrix}}_{\text{Grand average}}
$$

$$
+
\underbrace{\begin{pmatrix} -1.75 & 11.25 & 2.25 & -11.75 \\ -1.75 & 11.25 & 2.25 & -11.75 \\ -1.75 & \boxed{11.25} & 2.25 & -11.75 \\ -1.75 & 11.25 & 2.25 & -11.75 \\ -1.75 & 11.25 & 2.25 & -11.75 \\ -1.75 & 11.25 & 2.25 & -11.75 \end{pmatrix}}_{\text{Treatment effect}}
+
\underbrace{\begin{pmatrix} -8.0 & -7.0 & -1.0 & -7.0 \\ 0.0 & 6.0 & 17.0 & 4.0 \\ -4.0 & \boxed{12.0} & 2.0 & -13.0 \\ 5.0 & -3.0 & -5.0 & 2.0 \\ -16.0 & 0.0 & -13.0 & 8.0 \\ 23.0 & -8.0 & 0.0 & 6.0 \end{pmatrix}}_{\text{Residual error}}
$$

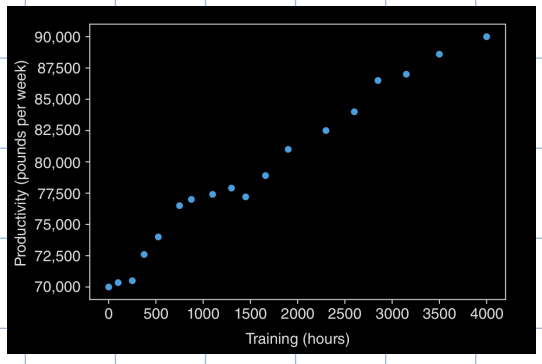
$$y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

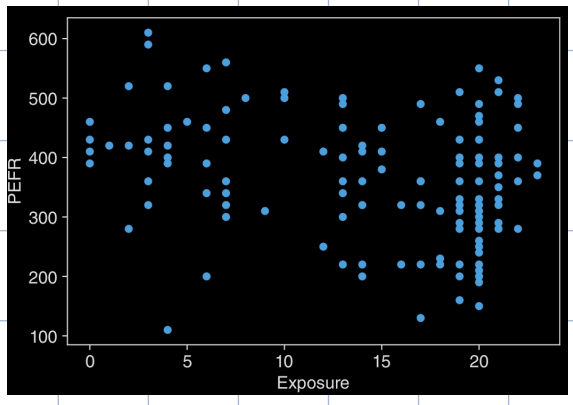
Chapter 9 Correlation

$(X_i, Y_i) \quad i = 1, 2, \ldots n$

Table 9.1 Training and productivity at Delta Wire.

Total Training Hours Cumulative	Productivity Pounds per Week
0	70,000
100	70,350
250	70,500
375	72,600
525	74,000
750	76,500
875	77,000
1100	77,400
1300	77,900
1450	77,200
1660	78,900
1900	81,000
2300	82,500
2600	84,000
2850	86,500
3150	87,000
3500	88,600
4000	90,000





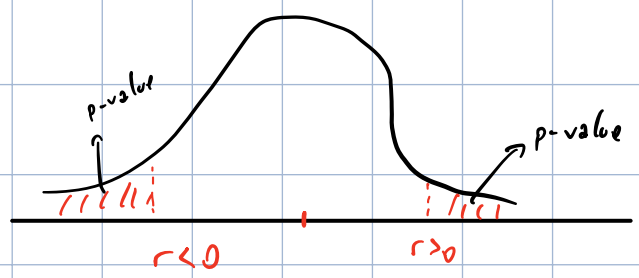
Pearson Correlation Coefficient : $-1 \leq r = \dfrac{cov(X,Y)}{s_x \cdot s_y} = \dfrac{\sum_{i=1}^{n} (x_i - \bar{x}).(y_i - \bar{y})}{(n-1).s_x.s_y} \leq 1$

X has $\bar{X}$, s_x
Y has $\bar{Y}$, s_y

correlation is between -1 and 1.
if ≥ 0, positive correlation, else negative correlation.

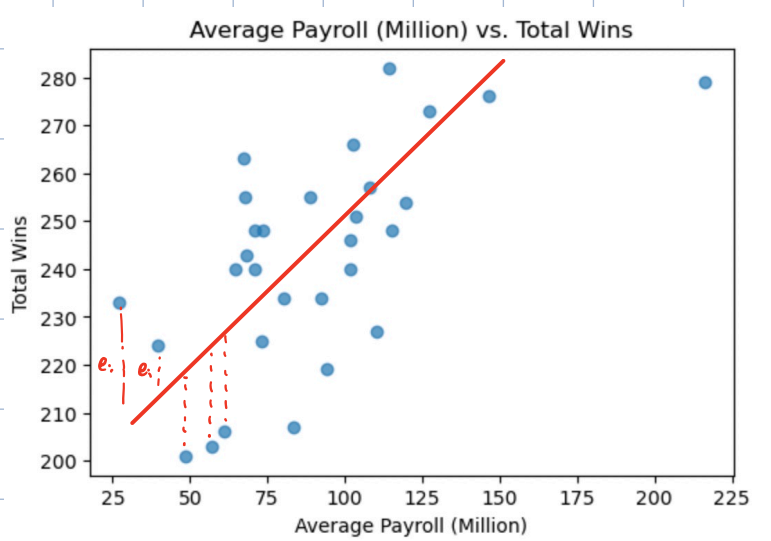
H_0: There is absolutely no association between X and Y.

- Find a p-value. (Permutation test)



If there is correlation, then

$y = ax+b$



$y = \beta_0 + \beta_1 x$

$\hat{y} = \beta_0 + \beta_1 x$

(x_i, y_i) , $\hat{y}_i = \beta_0 + \beta_1 x_i$ $\quad e_i = y_i - \hat{y}_i$

$S(\beta_0, \beta_1) = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$

Sum of Squared Errors

Goal: $\min_{\beta_0, \beta_1} S(\beta_0, \beta_1)$ unconstrained optimization

$$\frac{\partial S}{\partial \beta_0} = 0 \quad , \quad \frac{\partial S}{\partial \beta_1} = 0$$

$$\frac{\partial S}{\partial \beta_0} = \sum_{i=1}^{n} (-2).(y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\sum_{i=1}^{n} y_i - \sum_{i=1}^{n} \beta_0 - \beta_1 \sum_{i=1}^{n} x_i = 0$$

$$\frac{1}{n} \sum_{i=1}^{n} y_i - \beta_0 - \beta_1 . \frac{1}{n} . \sum_{i=1}^{n} x_i = 0$$

$$\bar{y} - \beta_0 - \beta_1 \bar{x} = 0$$

$$\boxed{\beta_0 = \bar{y} - \beta_1 \bar{x}}$$

$$\frac{\partial S}{\partial \beta_1} = \sum_{i=1}^{n} (-2 x_i).(y_i - \beta_0 - \beta_1 x_i) = 0$$

$$= \sum_{i=1}^{n} (y_i x_i - \underbrace{x_i \beta_0}_{\text{insert } \beta_0} - \beta_1 x_i^2) = 0$$

$$= \sum_{i=1}^{n} (y_i x_i - (\bar{y} - \beta_1 \bar{x}) x_i - \beta_1 x_i^2) = 0$$

$$= \sum_{i=1}^{n} y_i x_i - \bar{y} \underbrace{\sum_{i=1}^{n} x_i}_{n.\bar{x}} + \beta_1 \bar{x} \underbrace{\sum_{i=1}^{n} x_i}_{n.\bar{x}} - \beta_1 \sum_{i=1}^{n} x_i^2$$

$$= \sum_{i=1}^{n} y_i x_i - n.\bar{x}\bar{y} + \beta_1 . n.\bar{x}^2 - \beta_1 \sum_{i=1}^{n} x_i^2 = 0$$

$$\sum_{i=1}^{n} Y_i X_i - n.\bar{X}.\bar{Y} = \beta_1 . \left(\sum_{i=1}^{n} X_i^2 - n.\bar{X}^2 \right)$$

$$\beta_1 = \frac{\sum_{i=1}^{n} Y_i X_i - n.\bar{X}.\bar{Y}}{\sum_{i=1}^{n} X_i^2 - n.\bar{X}^2} = \frac{\sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n} (X_i - \bar{X})^2} \qquad \boxed{\beta_1 = \frac{cov(X,Y)}{Var(X)}}$$

$$\boxed{r = \frac{cov(X,Y)}{S_X . S_Y}} \qquad \boxed{\beta_1 = r . \frac{S_Y}{S_X}}$$

$$SST = \sum_{i=1}^{n} (Y_i - \bar{Y})^2 \leftarrow var(Y).(n-1)$$ (mean Y'dan farkı)

total sum of squares

$$SSR = \sum_{i=1}^{n} (\hat{Y}_i - \bar{Y})^2$$ (regression line'ın mean'den farkı)

Sum of squares of regression

$$SSE = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^{n} e_i^2$$

Sum of squared errors

$$SST = \underbrace{SSR}_{\text{explained by regression}} + \underbrace{SSE}_{\text{unexplained part}}$$

total

Define $R^2 = \frac{SSR}{SST}$ ratio of the explained variance

For single variable linear regression, $R^2 = r^2$

$$0 \leq R^2 \leq 1$$

$$MEDV = \beta_0 + \beta_1 . CRIM + \beta_2 . RM$$

[8]:
```
import statsmodels.api as sm

df.head()
X = df[["CRIM","RM"]] #inputs
y = df["MEDV"] #output
```

```python
import statsmodels.api as sm

X = df[["CRIM","RM"]] #inputs
y = df["MEDV"] #output
X = sm.add_constant(X) #for adding \beta_0

model = sm.OLS(y,X).fit()
predictions = model.predict(X)
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                   MEDV   R-squared:                       0.449
Model:                            OLS   Adj. R-squared:                  0.445
Method:                 Least Squares   F-statistic:                     107.8
Date:                Tue, 05 May 2026   Prob (F-statistic):           6.03e-35
Time:                        09:11:11   Log-Likelihood:                -712.99
No. Observations:                 267   AIC:                             1432.
Df Residuals:                     264   BIC:                             1443.
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      β0  2.5088      2.822      0.889      0.375      -3.047       8.064
CRIM       β1 -0.4904      0.040    -12.303      0.000      -0.569      -0.412
RM         β2  3.0083      0.461      6.531      0.000       2.101       3.915
==============================================================================
Omnibus:                        7.375   Durbin-Watson:                   1.883
Prob(Omnibus):                  0.025   Jarque-Bera (JB):               12.275
Skew:                           0.044   Prob(JB):                      0.00216
Kurtosis:                       4.047   Cond. No.                         101.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Proportion of variance explained by model

$H_0: \beta_0, \beta_1, \beta_2 \ldots = 0$

p-value

degree of freedom

p-value

Adding new variable (CRIM_RM)

```python
df["CRIM_RM"] = df["CRIM"] * df["RM"]
X_int = sm.add_constant(df[["CRIM", "RM","CRIM_RM"]])
model_int = sm.OLS(y,X_int).fit()
print(model_int.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                   MEDV   R-squared:                       0.491
Model:                            OLS   Adj. R-squared:                  0.485
Method:                 Least Squares   F-statistic:                     84.43
Date:                Tue, 05 May 2026   Prob (F-statistic):           2.74e-38
Time:                        09:26:13   Log-Likelihood:                -702.63
No. Observations:                 267   AIC:                             1413.
Df Residuals:                     263   BIC:                             1428.
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         -7.6236      3.497     -2.180      0.030     -14.510      -0.737
CRIM           1.2469      0.379      3.290      0.001       0.501       1.993
RM             4.6799      0.573      8.163      0.000       3.551       5.809
CRIM_RM       -0.2916      0.063     -4.607      0.000      -0.416      -0.167
==============================================================================
Omnibus:                       16.819   Durbin-Watson:                   1.860
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               38.218
Skew:                           0.251   Prob(JB):                     5.02e-09
Kurtosis:                       4.784   Cond. No.                         650.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

decreased

boston housing data ile ödev en iyi regressionu bul. kendi kolonumu ekleyebiliriz burada ayrıca variable çıkarılınca adj. R-sq büyüyor daha fazla eklemek

Multi-Variable

OLS : Ordinary Least Squares

$(x_1^1, \dots, x_k^1, y^1), \dots\dots, (x_1^n, \dots, x_k^n, y^n)$ $\quad$ k features, n data points

not exact solution, but a close prediction

regression model $\hat{y}^i = \hat{\beta}_0 + \hat{\beta}_1 x_1^i + \dots + \hat{\beta}_m x_m^i = (X^i)^T \cdot \hat{\beta}$

$$e^i = y^i - \hat{y}^i$$

$$\underbrace{\underset{\beta_0 \dots \beta_m}{\text{minimize}}}_{\beta} \sum_{i=1}^{n} (e^i)^2 = \min_{\beta} \sum_{i=1}^{n} (y^i - \hat{y}^i)^2 = \sum_{i=1}^{n} (y^i - (x^i)^T \cdot \hat{\beta})^2$$

vectors

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}_{(k+1)\times 1} \qquad X^i = \begin{bmatrix} 1 \\ x_1^i \\ x_2^i \\ \vdots \\ x_k^i \end{bmatrix}_{(k+1)\times 1}$$

Define $$X = \begin{bmatrix} 1 & x_1^1 & \dots & x_k^1 \\ \vdots & & & \vdots \\ 1 & x_1^n & \dots & x_k^n \end{bmatrix}_{n\times(k+1)} \qquad Y = \begin{bmatrix} y^1 \\ \vdots \\ y^n \end{bmatrix}_{n\times 1}$$

$$\sum_{i=1}^{n} (e^i)^2 = (\underbrace{Y}_{n\times 1} - \underbrace{X\beta}_{n\times 1})^T \cdot (Y - X\beta)$$

$(1\times n) \cdot (n\times 1) : (1\times 1)$

$$\min_{\beta} (Y - X\beta)^T \cdot (Y - X\beta) = f(\beta)$$

by changing β

$$\begin{bmatrix} \frac{\partial f(\beta)}{\partial \beta_0} \\ \vdots \\ \frac{\partial f(\beta)}{\partial \beta_m} \end{bmatrix} = 0$$

∇f : gradient of f

set $\nabla f = 0$

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \epsilon$$

If errors satisfy the assumptions (else bootstrapping)

$$s^2 = \frac{\sum_{i=1}^{n} \epsilon_i^2}{n - (k+1)}$$ estimated variance

$var(\hat{\beta}) = s^2 (X^T X)^{-1}$ $\longrightarrow$ result is a square matrix

$$\begin{bmatrix} \ddots & & \\ & \ddots & \\ & & \ddots \end{bmatrix} \rightarrow \text{covariances}$$

standard error

$se(\hat{\beta}_j) = \sqrt{var(\hat{\beta})_{jj}}$

$CI = \hat{\beta} \pm t_{\alpha/2, n-(k+1)} \cdot se(\hat{\beta})$

t kullanmamızın nedeni estimated variance kullanmamız. variance'ı bilseydik z kullanırdık.

kağıt tarafında iki variable ile regression, matris tersi alma vs.

sonra kod tarafında doğrulama